Ex: $L = \{a^n b^n c^n \mid n \geq 0\}$ is not a CFL.

Assume L is a CFL.

Then L must follow pumping lemma.
Suppose pumping length is p.

$S = a^p b^p c^p$, $|S| = 3p \geq p$

Then by pumping lemma, s must be able to be decomposed into 5 components where $s = uvxyz$, s.t

1. $uv^i x y^i z \in L$, $\forall i \geq 0$
2. $|vy| > 0$
3. $|vxy| \leq p$

$$S = \underbrace{aaa \ldots\ldots a}_{p \text{ \# of a's}} \underbrace{bbb \ldots\ldots b}_{p \text{ \# of b's}} \underbrace{ccc \ldots\ldots c}_{p \text{ \# of c's}}$$

case 1: vxy contains the same set of characters. Then v and y can only be allocated to the same type of character. Therefore, when we pump uv^ixy^iz, the # of one type of characters increases compared to other two types of characters.

$\therefore uv^ixy^iz \notin L$

case 2: vxy contains mix of characters.

subcase 2.1, v and y contain the same character. We handled this in case 1.

ex: $vxy = aaa\ldots a\,bbb\ldots b$

$v = aaa\ldots a \quad y = aaa\ldots a$

subcase 2.2 v and y contain two different character types.

ex: $vxy = bbb\ldots b\,ccc\ldots c$

$v = bbb\ldots b \quad y = ccc\ldots c$

Now, when we pump up, uv^ixy^iz contains different number of type 3 character.

$\therefore uv^ixy^iz \notin L$

Subcase 2.3 either v or y contains a mix of characters.

$$vxy = aaa \cdots a\, bbb \cdots b$$

$$v = aa \cdots a\, b \cdots b \qquad y = bb \cdots b$$

Now, when we pump up, uv^ixy^iz contains different order of a, b, and c's.

$\therefore uv^ixy^iz \notin L$

Now, we considered all the possible decompositions for s, and none of them are pumpable.

$\therefore L$ is not a CFL.

$C = \{ ww \mid w \in \{0,1\}^* \}$ is not a CFL.

Assume C is a CFL.
Then C must follow pumping Lemma.
Suppose pumping length is p.

Let $S = 0^p 1 0^p 1$, $|S| = 2p+2$

$$S = \underbrace{\overbrace{000\ldots0}^{p\# 0's}}_{u}\ \underbrace{0\ 1\ 0}_{v\,x\,y}\ \underbrace{\overbrace{00\ldots0}^{p\# 0's}\,1}_{z}$$

$$uv^ixy^iz = 0^{p-1}0^i 1 0^i 0^{p-1} 1$$

$$= 0^{p-1+i} 1 0^{p-1+i} 1$$

This String does not work.

Pick $S = 0^p 1^p 0^p 1^p$, $|S| = 4p$

$$S = 000\ldots\underbrace{0111}_{vxy}\ldots1 \mid 000\ldots0111\ldots1$$

$$|uv^2xy^2z| = 4p + |vy|$$

middle point would be

$$2p + \frac{|vy|}{2}$$

Since $|s| \geq p$, s must be able to be decomposed into $s = uvxyz$ s.t

1. $uv^i xy^i z \in C$, $\forall i \geq 0$
2. $|vy| > 0$
3. $|vxy| \leq p$

case 1: vxy does not straddle the mid point of s.

In this case $uv^2xy^2z \notin C$.

The reason is that, in uv^2xy^2z, the first character of string 1 and first character of string 2 does not match.

case 2: If vxy cross mid point, consider $uv^0xy^0z = uxz = 0^p1^i0^j1^p$, where $i, j \neq p$.

$\therefore uv^0xy^0z \notin C$.

$\therefore C$ is not a CFL.

Prove $L = \{a^n b^j \mid n = j^2\}$ is not a CFL.

Assume L is a CFL

Then let p be the pumping length

Let $s = a^{p^2} b^{p}$

breakpoint.

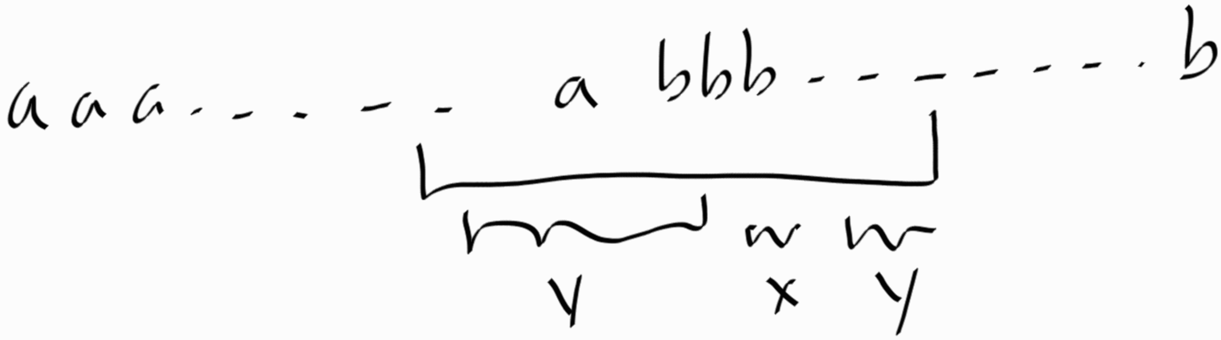
$$s = \underbrace{aaaa\cdots\cdots a}_{p^2 \text{ \# of } a\text{'s}}\ \underbrace{bbbb\cdots\cdots b}_{p \text{ \# of } b\text{'s}}$$

Case 1: vxy does not cross the breakpoint.

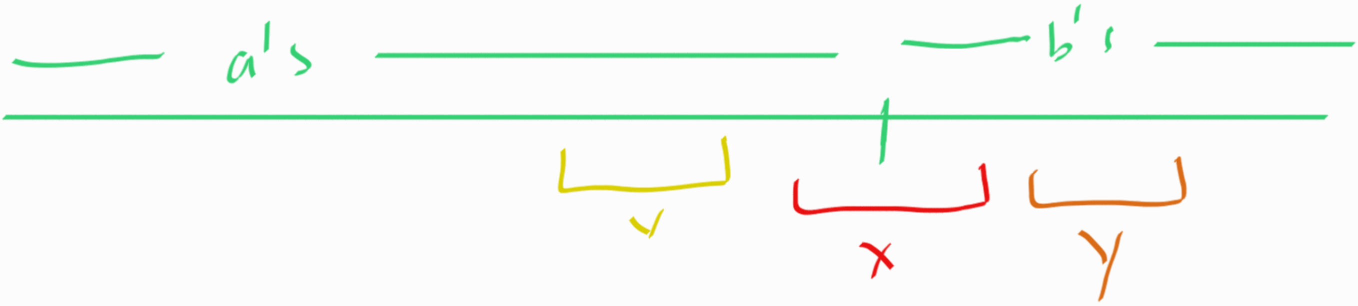
Then, when pumping up, we will have different # of a's and b's which would violate the $n=j^2$ condition.

Case 2: vxy crosses the midpoint.

case 2.1: v or y contains two types of characters. then when you pump up, we have out of order characters.



case 2.2 Both v and y does not cross the breakpoint



k_1 # of a's in v

k_2 # of b's in y

consider uv^ixy^iz

$|uv^ixy^iz| = p^2+p+k_1(i-1)+k_2(i-1)$

subcase 2.2.1 $(k_1=0)\wedge(k_2=0)$

cannot happen as it

violates condition 2

subcase 2.2.2 $(k_1=0)\vee(k_2=0)$

then $uv^0xy^0z \notin L$, since this would

imbalance the # of a's and b's.

$uv^0xy^0z \notin L$

subcase 2.2.3

$(k_1 \neq 0)\wedge(k_2 \neq 0)$

$|vxy| = k_1+k_2+|x|$ —— (1)

uv^ixy^iz, and pumpdown

$|uv^0xy^0z| = p^2+p-k_1-k_2$

suppose $uxz \in L$

$p^2-k_1 = (p-k_2)^2$

$$\cancel{p^2} - k_1 = \cancel{p^2} - 2pk_2 + k_2^2$$

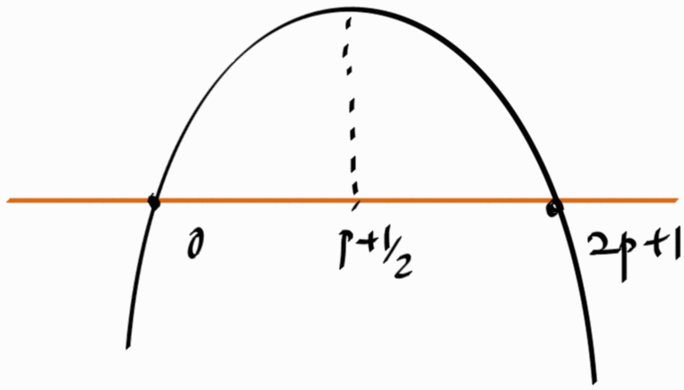
$$k_1 = -k_2^2 + 2pk_2$$

consider ①

$$|vxy| = k_1 + k_2 + |x|$$

$$= \underbrace{-k_2^2 + 2pk_2 + k_2}_{f(k_2)} + |x|$$

$$f(k_2) = -k_2^2 + k_2(2p+1) = -k_2(k_2 - (2p+1))$$



$$f'(k_2) = -2k_2 + 2p+1$$

$$f'(k_2) = 0 \Rightarrow k_2 = p + 1/2$$

$f(k_2)$ increases from 1 to p

$f(1) = -1 + 2p + 1 = 2p$

$f(p) = -p^2 + p(2p+1) = p^2 + p$

Here, we show that $f(k_2)$ which is the length of $|vy|$,

$\therefore \forall k_2 \in [1 \dots p] : f(k_2) > p$

But $|vxy| \geq f(k_2) > p$

$\therefore \forall k_2 \in [1 \dots p] : |vxy| \geq p$

what this means is for uxz to be in L any decomposition, we must have $|vxy| \geq p$, but this would violate condition 3 of pumping Lemma.

L is not a CFL.